MAINTENANCE MANUAL FOR INOPERATIVE ELEVON ON ECOSOAR FLYING WING UAV

Elevon on the Ecosoar stops working. There are several potential causes for an inoperative elevon on the Ecosoar:

A. Improper channel mapping:
The control channel may not be correctly mapped in QGroundControl or Mission Planner.

B. Intermittent or failed connections:
Servo connectors at the flight controller or along the servo extension (jumper) harness may have a loose or broken contact.

C. Mechanical failure of the servo:
The servo motor may have internal gear damage, preventing torque transfer to the output shaft.

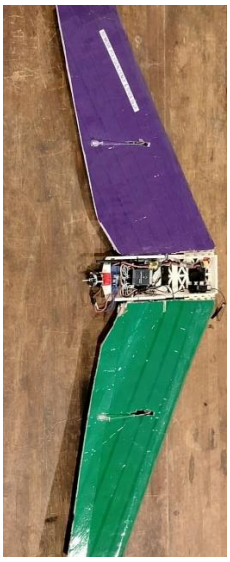
To isolate the root cause, first verify whether the servo produces any sound when a control input is commanded from the radio transmitter. If the servo makes noise but does not move the pushrod/elevon: The internal gear train is likely stripped or damaged, and the servo must be replaced. The following procedure describes how to safely replace a failed elevon servo.

1. Make the aircraft ready for maintenance

i. Remove the avionics bay top cover and disconnect the main battery to ensure the system is fully de-energized (see image 1a).
ii. Remove the propeller from the motor using a 1 mm Allen key inserted into the prop hub; rotate counter-clockwise to loosen and detach the propeller.
iii. Place the aircraft flat on its belly to avoid unnecessary bending or torsional loads on the structure during maintenance (refer to image 1c).
iv. Because the UAV uses winglets, it cannot rest flat on its belly with them installed.
v. Remove the winglets on both sides using a 2 mm Allen driver (refer to image 1b).

1a. Avionics bay

1b. Winglets section

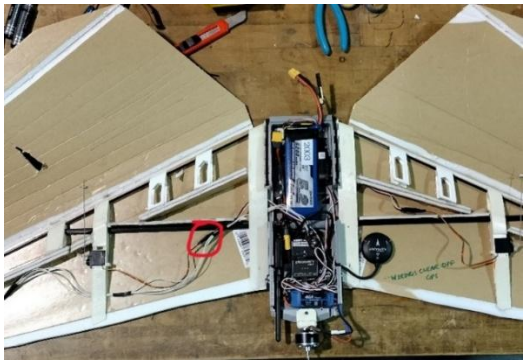

1c. UAV without winglets

2. Open the wing to access the servo motor

i. Before opening the wing, detach the servo arm from the control horn.
ii. Unscrew the linkage stopper as shown in image 2a to free the pushrod from the control horn.
iii. The top wing panel is hot-glued to the bottom panel.
iv. Use utility knife in combination with a hot glue gun without a glue stick to carefully soften and cut through the glue bond, separating the top and bottom panels (refer to image 2b).
v. Work gradually along the seam to avoid damaging the foam structure.
vi. Once the interior is accessible, disconnect the servo lead from the servo extension (jumper) cable as shown in image 2c.
vii. Using a scraper or utility knife, carefully break the glue bond and remove the faulty servo from the bottom wing panel (image 2d).
viii. Take care not to damage the surrounding foam ribs, spars, or wiring.

2a. Unscrew linkage stopper

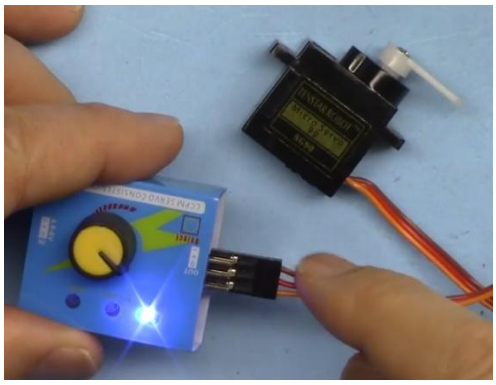
2b. Use utility knife & glue gun to open wing



2c. Disconnect servo from jumper extension

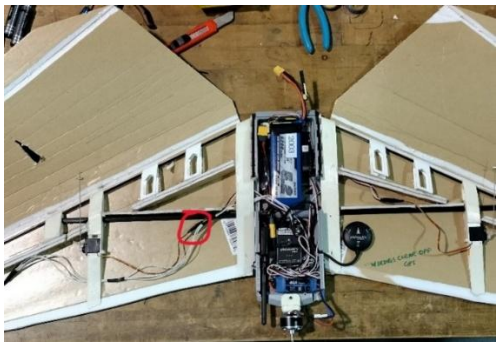
2d. Remove bad servo motor

3. Replace the servo motor
   i. Use a **servo tester** to verify that the new metal-gear servo operates smoothly across its full travel range and centers correctly (image 3a).
   ii. Once the servo passes the functional check, use a hot glue gun to bond it to the bottom wing panel in the original mounting location (image 3b).
   iii. Ensure the servo is aligned so that its neutral position corresponds to the elevon's neutral (trim) position.
   iv. Attach the servo connector to the jumper extension cable (image 3c), ensuring correct orientation of signal, power, and ground.
   v. Place the top wing panel back onto the bottom panel **without gluing** at this stage.
   vi. Reattach the servo control arm to the control horn and reconnect the battery.
   vii. Using radio input, verify that:
      - The new servo moves smoothly and without jitter.
      - The elevon deflection direction and magnitude match the opposite elevon (image 3d).

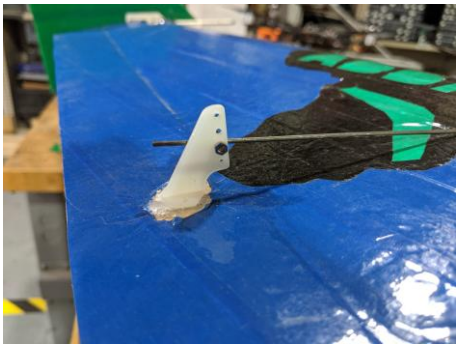


3a. Servo tester to check servo motor

3b. Stick the servo with help of glue gun



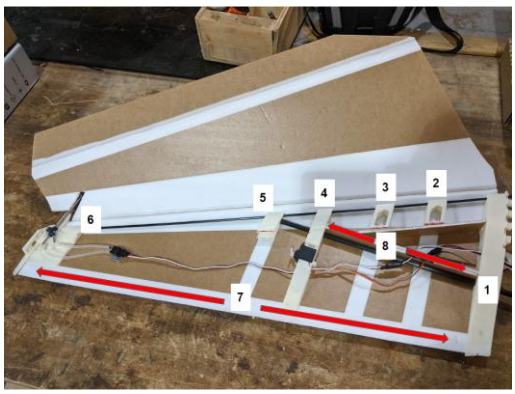
3c. Connect servo to jumper extension



3d. Attach control arm to control horn

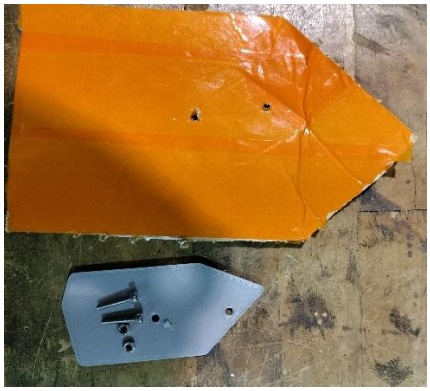
4. Close the wing and perform final functional checks

   i. Disconnect the battery again before gluing.
   ii. Apply hot glue along the designated contact points (1 through 8) as shown in image 4a.
   iii. Close the wing by pressing the panels together in the same sequence (1 through 8) to ensure even bonding and alignment.
   iv. Hold or support the wing until the hot glue has fully cooled and set.
   v. Reconnect the battery and perform a full radio check.
   vi. Confirm that both left and right elevons respond correctly, symmetrically, and without binding across the full control range.
   vii. Attach the winglet on each side and happy flying (refer image 4c and 4d).

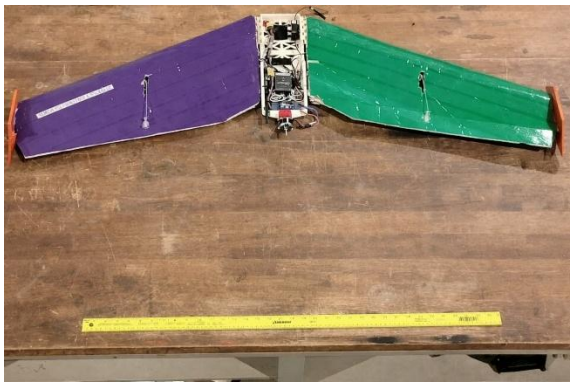


4a. Glue and pressure points on wing

4b. Closed wing



4c. Winglets with fasteners



4d. Assembled Ecosoar UAV